

Weather System Impact — 8v8 Density Sim

Tempest Fauna Trail

Generated by `tools/simulation/weather_impact.py`

2026-06-04

1 Question

How much do the two weather systems actually swing combat at large team sizes, as a function of *how many champions are drawn for the live weather affinity*? The two systems are decoupled in `src/game/weather_effects.py`:

- **System A — Weather Favor** (`combat_modifier`): the node weather buffs/debuffs each piece by its affinity at combat init. *“Who thrives under these conditions?”*
- **System B — Affinity Clash** (`damage_modifier`): a per-hit attacker-vs-defender multiplier on the predator/prey ring, weather-independent. *“Do I beat this enemy?”*

2 Method

Teams are size 8. Both sides always draw the *same multiset of tiers*, so the power budget is equal by construction (power depends only on tier and level); only the affinity filling each tier-slot varies. **CLEAR** affinity is inert in *both* systems (identity Favor, 1.0 Clash), so it is the isolator and filler.

Metric. An 8-piece team spans 8 distinct tiers, so its top piece is $\sim 5\times$ the bottom ($2^{(T-1)/3}$); binary win/loss is then dominated by the strongest piece’s duel and is high variance. The primary read is therefore the continuous **HP margin** $m = (\text{hp}_A - \text{hp}_B) / (\text{hp}_A + \text{hp}_B) \in [-1, 1]$: every piece’s weather buff feeds it, so the systematic effect surfaces as a low-variance mean shift. Win% is reported alongside.

Side-bias control. Each matchup is also played with sides swapped and folded (margin negated, outcome flipped). This cancels a combat-engine input-order advantage (Section 8); with it on, identical mirror cells read exactly 0.000 margin / 50%.

Engine is byte-deterministic; variance comes only from sampling many tier-windows (seeded). Runs below: System A/B/AB matrices at 30 samples/cell, density curves at 40 samples/cell, side-swap on.

3 System A — Weather Favor vs own-weather density

Player fields k own-weather (rain) champs + clear fillers against an all-clear (inert) enemy under rain. The control runs the *same* teams under **CLEAR** weather (Favor off); the delta strips champ-design bias, leaving pure Favor.

k	own%	m_{favor}	m_{ctrl}	Δm	favor%	ctrl%	Δwin
0	0%	+0.000	+0.000	+0.000	50.0	50.0	+0.0
1	12%	+0.183	-0.100	+0.283	60.0	45.0	+15.0
2	25%	+0.300	+0.000	+0.300	65.0	50.0	+15.0
3	38%	+0.267	-0.250	+0.517	63.3	36.7	+26.7
4	50%	+0.133	-0.217	+0.350	56.7	40.0	+16.7
5	62%	+0.583	-0.033	+0.617	78.3	48.3	+30.0
6	75%	+0.350	-0.167	+0.517	68.3	43.3	+25.0
7	88%	+0.567	-0.283	+0.850	80.0	36.7	+43.3
8	100%	+0.383	-0.600	+0.983	68.3	20.0	+48.3

Read. Full-team Favor swing **+0.98** margin / **+48 pp** win (± 0.12 sem). Monotonic in Δm . Favor is a strong, steadily-accumulating nudge — each on-weather piece adds margin.

4 System B — Affinity Clash

CLEAR weather (Favor off); mono-affinity vs mono-affinity. Only the per-hit ring multiplier differs. Cell = player mean HP margin.

player \ enemy	mist	cloudy	rain	snow	thunder
mist	0.000	-0.967	-0.967	-0.083	+1.000
cloudy	+1.000	0.000	-1.000	-1.000	+0.867
rain	+0.967	+1.000	0.000	-1.000	-0.467
snow	-0.017	+1.000	+1.000	0.000	-0.733
thunder	-1.000	-0.833	+0.450	+0.900	0.000

By ring relation (player vs enemy), aggregated:

relation	mean margin	win%
primary_predator	+0.980	99.0
secondary_predator	+0.640	81.7
secondary_preym	-0.657	17.3
primary_preym	-0.940	3.7

Read. At mono-stack saturation, Clash is brutal: a primary-predator team wins 99%, a primary-prey team 4%. Antisymmetric ($m(X, Y) \approx -m(Y, X)$), mirror diagonal exactly 0.

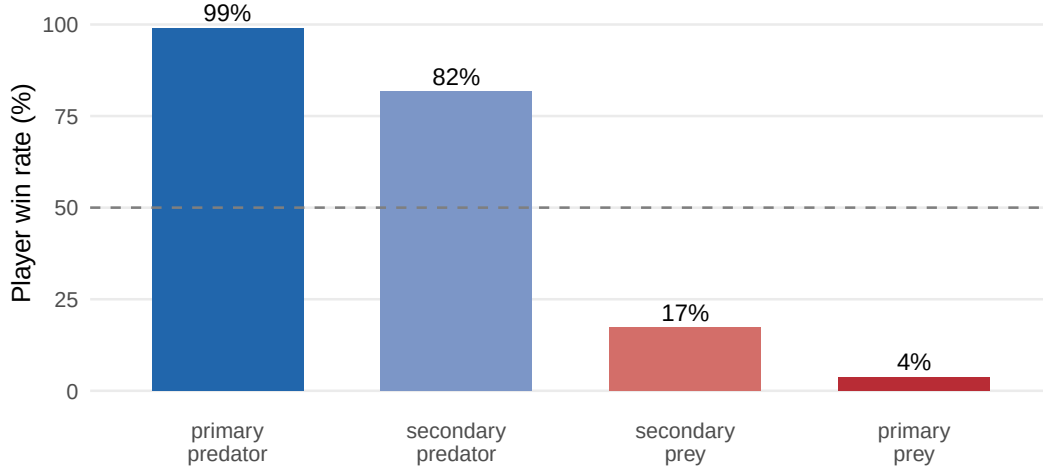


Figure 1: Affinity Clash at mono-stack saturation: player win rate by ring relation to the enemy. A wrong-side primary-prey matchup is near-unwinnable irrespective of weather.

Clash intensity vs predator-piece density

Player ramps j predator pieces + clear filler against a mono-prey enemy, CLEAR weather, pooled over all 5 ring weathers ($n = 400$ at $j=8$).

j	own%	margin	win%	Δm vs $j=0$	Δwin
0	0%	+0.482	74.0	+0.000	+0.0
1	12%	+0.627	81.2	+0.145	+7.3
2	25%	+0.795	89.8	+0.313	+15.7
3	38%	+0.795	89.8	+0.313	+15.7
4	50%	+0.858	92.8	+0.375	+18.8
5	62%	+0.922	96.0	+0.440	+22.0
6	75%	+0.950	97.2	+0.467	+23.3
7	88%	+0.980	99.0	+0.497	+25.0
8	100%	+0.990	99.5	+0.508	+25.5

Caveat. The $j=0$ anchor is 74%, not 50%: at $j=0$ this is clear-roster vs ring-roster (same tiers, different identities) and clear champions are statlined stronger on average — a separate roster-balance signal. The Δm column isolates the clash contribution from that bias.

5 System AB — both systems live (overlap)

Node weather = player affinity. Player gets Favor *and* its Clash vs the enemy. Cell = player mean HP margin.

weather \ enemy	mist	cloudy	rain	snow	thunder
mist	0.000	-1.000	-0.883	+0.433	+1.000
cloudy	+1.000	0.000	-1.000	-0.750	+1.000
rain	+1.000	+1.000	0.000	-1.000	+0.100
snow	+0.467	+1.000	+1.000	0.000	-0.783
thunder	-1.000	-0.400	+0.967	+1.000	0.000

Favor+Clash intensity vs own-affinity density

Player ramps j own-weather pieces + clear filler vs mono-prey, under weather X , pooled over 5 weathers.

j	own%	margin	win%	Δm vs $j=0$	Δwin
0	0%	+0.667	83.0	+0.000	+0.0
1	12%	+0.843	92.2	+0.175	+9.3
2	25%	+0.940	97.0	+0.272	+14.0
3	38%	+0.968	98.2	+0.300	+15.3
4	50%	+0.995	99.8	+0.328	+16.8
5	62%	+1.000	100.0	+0.333	+17.0
6–8	$\geq 75\%$	+1.000	100.0	+0.333	+17.0

Read. The systems compound: full mono saturates to 100% win by only $\sim 5/8$ on-affinity pieces. Crucially **Clash dominates Favor**: rain under rain still *loses* 0% to snow (snow preys rain) despite the Favor buff; Favor only fully cancels a *secondary*-prey clash gap (rain vs thunder $\approx 57\%$).

6 Cross-system: intensity vs density

The marginal value of each on-affinity piece, per mechanism. Favor (A) is measured vs a neutral clear enemy (control-subtracted); Clash (B) and the compound (AB) vs a mono-prey enemy (vs the $j=0$ anchor).

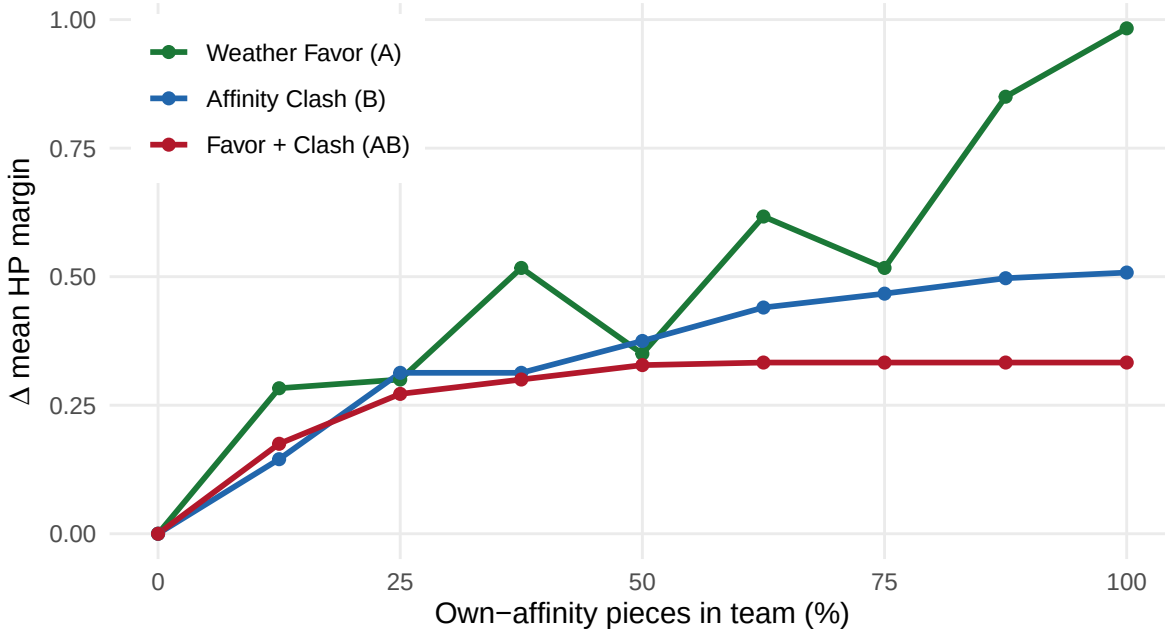


Figure 2: Δ mean HP margin as own-affinity density rises 0–100%. Read each curve’s *shape*, not cross-curve height: each is measured against a different baseline (Favor vs a 50% neutral enemy with full headroom; Clash and compound vs a high prey-baseline, so they are ceiling-compressed). All grow monotonically; Clash is the smoothest (low variance), Favor the noisiest (a single dominant top-tier piece still swings the board, motivating the margin metric).

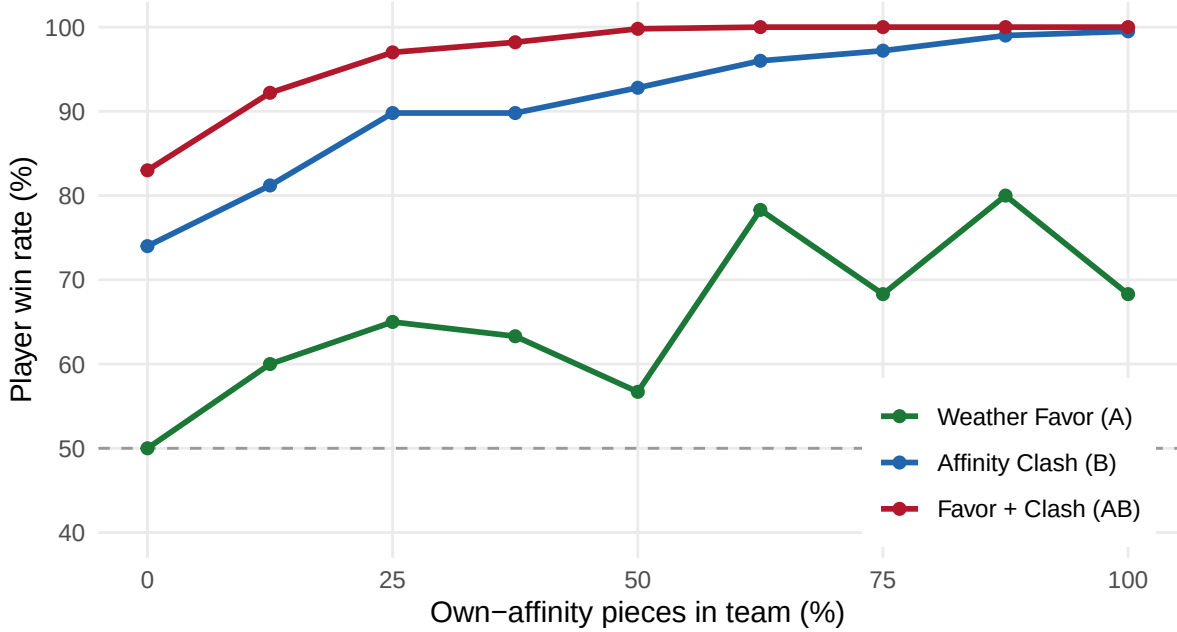


Figure 3: Absolute player win rate vs density. The compound (AB) reaches 100% at $\sim 5/8$ pieces. Favor’s win% (A) is noisier and lower-ceilinged than its margin suggests — a single dominant top-tier piece can still flip a high-margin board, which is exactly why margin is the primary metric.

7 Verdict

- Weather matters a lot. Both systems produce large, monotonic swings with own-affinity density.
- **Affinity Clash is the harder lever** — decisive even against an opposing Favor buff; a wrong-side primary-prey matchup is near-unwinnable ($\sim 4\%$) regardless of weather.
- **Weather Favor is a strong but overcomable nudge** (+48 *pp* at full stack) — it cancels a secondary clash gap but not a primary one.
- Mono-affinity stacking is overwhelming: ceiling at half the team.

8 Side-effect: a combat-engine bug

The mirror cells reading $\neq 50\%$ before side-swapping exposed a real engine bug. `resolve_combat` (`combat/legacy.py:469`) sets `speed_tiebreaker = index` over a team-block-then-enemy-block ordering, and `_event_sort_key` (`combat/loop_new.py:378`) breaks equal `attack_speed` ties by that index. So the entire team out-prioritises the entire enemy block on any AS tie — acting, hitting and killing first. An identical mirror is a *deterministic* side-A win, not a draw. Design intent is side-independent ordering (first mover by AS and MS only).

This biases every prior combat result and sim report. It is mitigated in this sim via `--both-sides` but is *not* fixed in the engine — a fix changes the determinism of all existing tests and is a pending decision. Candidate fixes: interleave tiebreakers (team0, enemy0, team1, ...) or fold `move_speed` into the tiebreak.